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S.320 PILING

1. SCOPE OF WORK

The Bidder shall furnish all labour, material and equipment necessary to design, fabricate, deliver, install and test all piles as indicated in Part 4 Subsection 1.2 “Summary of Work”, and as specified herein and as shown on the approved Bidder’s drawings.

2. CODES AND STANDARDS

2.1. The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

2.2. The following codes and standards extracted from Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design, manufacture and testing of the work included in this Specification:

ASTM American Society for Testing and Materials
ACI American Concrete Institute
JIS Japanese Industrial Standards
AIJ Architectural Institute of JAPAN

3. SOIL DATA

3.1. The Bidder shall carry out soil investigation in accordance with the Technical Specification S240 and shall state it in his Bid Proposal. The investigation program shall be subject to the Owner’s approval.

3.2. The soil investigation data for the existing plant and Block 5 are referred to Book IV Part 6 Bid Drawings “Soil Investigation Data “as reference.

4. PILE TYPE

Pile type shall be the circular hollow pre-stressed concrete Piles. The Bidder shall submit with his Bid Proposal details of the proposed pile including assumed length of pile and outside diameter.

4.1. The Bidder, however, may propose other pile types together with all the stipulated requirements as alternative.

4.2. The Bidder shall be fully responsible for furnishing the diameter required for driving sound piles. Head of piles shall have sufficient strength to withstand hammering, and concrete pile strength shall also be sufficient to withstand ground pressure from driving adjacent piles.

5. MATERIALS

5.1 CIRCULAR HOLLOW PRESTRESSED CONCRETE PILES

5.1.1 The circular hollow pre-stressed concrete piles (PC pile) shall have minimum diameter of 7 mm PC wire and shall conform to the requirements of ACI 318 and ACI 543 or JIS A – 5373/2004 and JIS G -3536/1999 (or latest).

5.1.2 All piles shall be new and shall be made locally at the approved manufacturer's factory.

5.1.3 Piles shall be supplied free from harmful defects and good commercial finish with regard to smoothness and freedom from loose scale and rust.

5.1.4 The test certificate for material to be used for piles shall be submitted to the Owner prior to pile acceptance and manufacturing.

5.1.5 When requested by the Owner, the Bidder shall accompany the Owner to the place of the approved manufacture's factory to witness test and inspection carried out on the piles.

5.1.6 For each consignment of piles delivered to the Site, the Bidder shall submit to the Owner copies of the manufacturer's certificate.

5.2 CIRCULAR HOLLOW PRESTRESSED CONCRETE PILES DESIGN

5.2.1 The PC pile shall be of high quality pile (class A) with concrete cover of minimum 60 mm. The PC pile with minimum diameter of 600 mm shall be applied for the major structures, and 350 mm and/or 500 mm for other struc-

tures, and the respective PC piles shall have high quality and shall be approved by the Owner.

- 5.2.2 The Bidder shall submit calculation sheet for the pile design based on the allowable loads, including safety factors, considering the soil data and structural loading carried by the pile.
- 5.2.3 Concrete compressive strength shall be of minimum 5kN/cm² (500kg/cm²) by using cement Type V for location near sea and Type II for other area.
- 5.2.4 The total number of pre-stress strands may be varied by increasing the number or size of the pre-stress strands.
- 5.2.5 Pre-stressing reinforcement shall be of stress relieved steel wire conforming to JIS G-3536 and the pre-stressed steel tensile strength shall be of minimum 1420N/mm².
- 5.2.6 All other non-pre-stressing reinforcing shall be of deformed steel bars conforming to JIS-G-3112, ASTM –A615 grade 40. Spiral reinforcing confirm to the requirement of ASTM –A82.

5.3 JOINING

Splicing of piles shall be performed by pile manufacturer's procedure which shall be submitted to the Owner for approval. Pile splices shall be located at a minimum of 3 m below the cut-off elevation. Splice strength shall not be less than pile bending and tensile capacity

6. PILE HANDLING AND TRANSPORT

Length of concrete piles shall be determined as to allow proper handling and transportation of piles, and shall be subject the Owner's approval.

The piles produced at the approved factory shall be transformed and delivered to the Site free of any damage.

7. LINES AND GRADES

The Bidder shall lay out lines and grades and be fully responsible for the correctness of such lines and grades and for the proper execution of the work to such lines and grades.

All surveying shall be in accordance with Specification S-240 "Surveying and Setting Out" and Part 4, Section 4, Erection and Commissioning.

8. PILING EQUIPMENT, INSTALLATION AND WORKMANSHIP

8.1 DRIVING EQUIPMENT

8.1.1

The piling equipment shall be capable of driving the piles to the ultimate resistance of 2 times nominal compression capacity/design load capacity.

If the pile location is close to the existing structures, equipment or the boundary line of the Site, pile driving method applied shall have no effecting to them, for example pre auger method is one of recommendable method.

8.1.2 Hammer size and Cushion Block

8.1.2.1 Hammer size – the piles shall be hydraulic driven. The hammer shall develop the minimum energy per blow to achieve the required pile capacity or an equivalent set based upon the rated energy of the actual piling hammer used. The final set will be adjusted as necessary after driving and testing of the tests pile, the set will be valid only for the hammer used to drive the test piles.

8.1.2.2 Cushion blocks type for pile driving which has to provide sufficient function of noise reduction and preventing damage to pile head shall be submitted to the Owner for approval. Cap blocks shall be inspected periodically during driving and no driving shall be done with blocks that have been unduly worn and compacted with use.

8.1.2.3 Pile cushion blocks for all piles shall be of the same thickness, material, quality and construction. Uneven and over sized cushion caps shall not be allowed.

8.1.3 Pile drivers with firmly supported fixed leads extending down to the lowest point that the hammer must reach are acceptable, short leads suspended from a line and braced only by lines will not be acceptable unless pile are rigidly braced and held in alignment by suitable guide frames prior to pile driving. Leads with spear (auger) points fixed to the lower lead end, embedded in the ground adjacent to the pile and fixed or suspended from the pile driving rig are acceptable, provided the leads remain stationary during pile driving. At all stages during driving, the pile shall be supported and re-

strained by means of leaders, trestles, temporary supports or other guide arrangements to maintain position and alignment and to prevent buckling.

- 8.1.4 The piling equipment and driving force used shall ensure that the piles are installed without damage or loss of strength. If at any time in the opinion of the Owner satisfactory result cannot be obtained with the equipment furnished by the Bidder, equipment of other rated capacity capable of driving piles to the approved elevation shall be furnished and used by the Bidder and no additional cost to the Owner.

8.2 MARKING AND LENGTH OF PILES

8.2.1 Marking of Piles

Each pile shall be clearly identified by its number with white paint at least every three (3) meters, with a minimum of three locations at the top, bottom and middle. The length shall be marked and identified with white paint at least every meter intervals of 300 mm shall be also marked with paint.

8.2.2 Length of Piles

The length of the piles shall be as necessary to develop the required pile capacity. The Bidder shall determine length of piles based on geotechnical data and shall be proven by pile load tests.

8.3 SETTING OUT

Firstly, the main setting out for the piles shall be completed prior to commencement of piling. Secondary, individual pile setting out shall be completed and agreed upon not less than 8 hours prior to commencing work of the piles concerned. All main setting out points, lines, stations and other work shall be maintained safe and undisturbed.

8.4 DRIVING OF PILES

- 8.4.1 All piles shall be satisfied with the tolerances for position and inclination as stated herein. The deviation in position of the as-driven centerline of a pile or the projection of its centerline from the pile center point as shown on the setting out drawings shall not exceed 75 mm when measured in a horizontal direction at the defined level. Deviations from vertical shall not be more than 1 in 100 from the specified batter.
- 8.4.2 If piles are installed out of plant position or line beyond allowable tolerances, the piles shall be extracted or additional piles shall be driven, as requested by the Owner, the properly provide for load conditions and to avoid overstressing of piles. Manipulation of piles to force them into proper position or line will not be permitted. The Bidder shall be responsible to extract and/or drive additional piles as noted above and without additional cost to the Owner.
- 8.4.3 Damaged pile shall be abandoned, replaced or repaired as approved by the Owner at the Bidder's expense. Abandoned piles shall be filled with soil and without additional cost to the Owner. Additional piles shall be driven as requested to provide for load conditions and to avoid overstressing of piles.
- 8.4.4 Piles shall be driven until the required final set is achieved.
- 8.4.5 If files do not achieve the required final set at the specified depth, pile shall be driven deeper until specification is met and upper portion shall be extended by welding and without additional cost
- 8.4.6 Driving of piles shall not begin until the evaluation of the preliminary pile load tests for tension load, lateral load and compression load to failure has been completed for each diameter type and proposed for approval of the Owner.
- 8.4.7 The Bidder shall monitor vibration and ground movement during pile driving. A proposal of the monitoring method shall be submitted to the Owner for approval.
- 8.4.8 Build Ups
- 8.4.8.1 The Bidder shall submit build ups pile to the Owner for approval.

8.4.8.2 In the event that any pile is driven below the established cut off elevation or if pile heads are damaged during driving, all necessary build ups shall be provided.

8.4.8.3 Build ups shall consist of all required pipes, splices, concrete reinforcing steel, etc. required to bring the top of the pile to the cut off elevation. The cost of pile build ups shall be included in the Contract Price.

8.4.9 Pile Heaving

8.4.9.1 After piles have been driven, the Bidder shall check to determine if any piles heave due to driving of adjacent piles and re-drive that heaved piles as directed by the Owner and the cost of re-driving shall be included in the Contract Price.

8.5 TEMPORARY PROTECTION OF PILES

The Bidder shall ensure that all free-standing piles are temporary braced or stayed to prevent damage to the piles or loosening of the files in the supporting ground. The welding of temporary works, such as bracing or shuttering supports, to the permanent piles shall not be permitted. The Bidder shall be responsible for all such temporary works and for any repairs, and also for the replacement of any pile which is damaged in any way and without additional cost to the Owner.

8.6 PILE CUT-OFF

On completion of the driving of the piles to the levels approved by the Owner, the heads of all at elevations indicated only after there is no danger from heaving. All cut-offs shall become property of the Bidder and shall be removed by him from the Project Site.

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8.7 EXTRACTION OF PILES

When extraction of the piles is required, it shall be carried out using efficient extraction equipment of adequate capacity. The extraction equipment shall be correctly positioned over the pile to minimize damage to the pile head and to prevent any major disturbance of the supporting ground. Pile heads damaged during extraction and file toes found to have been damaged during initial driving shall be trimmed with a clean cut perpendicular to the pile axis prior to re-driving. Extension pieces shall be added as necessary to restore the pile to its original length and approved by the owner.

9. PILE TO PILE CAP CONNECTION

- 9.1. Adequate structural connection between the piles and the pile caps shall be provided to meet seismic loading design requirements.
- 9.2. Pipe shall be filled by gravely sand up to certain elevation or plywood end plate set to the adequate level for concreting and then filled by lean concrete as shown on the approved Bidder's drawing. Lean concrete as a base of reinforcing concrete of pile top. The pile walls shall be thoroughly cleaned of soil using an approval method before filled by concreting.
- 9.3. Do not place pile top concrete in any pile until the pipe and reinforcement as indicated in the approved Bidder's drawing has been inspected and agreed for concreting by the Owner.
- 9.4. Placing concrete in each shell or pipe shall be in such manner as to avoid segregation of materials and to prevent air entrapment which could cause voids. Concrete shall not be chute directly into the piles but shall be placed upper the gravel & plywood cover the top of piles.
- 9.5. Placing concrete to top of pile elevation shall be indicated on the approved Bidder's drawings.
- 9.6. All concrete work shall be in accordance with the requirements of Specification S-340 Concrete Works.

10. RECORDS OF PILE CONSTRUCTION

During the course of the work, the Bidder shall maintain a complete record of pile construction and the Bidder shall submit to the Owner, in an approved format, a record for each pile. Each pile record shall be submitted immediately after completion of each pile and shall include the followings:

- 10.1. Type of pile.
- 10.2. Pile number and all principal characteristics of the file including dimensions.
- 10.3. The depth and rate of penetration, in relation to site datum, for the length of the pile.
- 10.4. Details of driving and installation sequence progress, including dates, times, equipment used and personnel (operator, engineer, etc)
- 10.5. The equipment hammer: model number, weight and rated energy.
- 10.6. Number of hammer blows for each 300 mm of driving/penetration.
- 10.7. Actual location of piles driven, with variations from plan locations indicated.
- 10.8. Variations in plumb or batter.
- 10.9. Set on completion of the initial drive and final set on completion of the final set on completion of the final re-drive.
- 10.10. Any unusual phenomena encountered in driving piles.
- 10.11. Depth and type of soil strata.
- 10.12. Observations of ground water level and groundwater inflow.
- 10.13. Name of inspector for each pile.
- 10.14. Cushion material identification when changed, when material added.
- 10.15. Top elevation when driving is complete.
- 10.16. Cut-off elevation.
- 10.17. Depth of augering and auger size for pre-auger.
- 10.18. The record of pile construction shall be submitted in accordance with the requirement set forth in Part 4, Subsection 2.2.

11. PILE LOAD TESTING

11.1 SCOPE OF WORK

- 11.1.1 The work specified under this section includes furnishing all labour, materials and equipment for compression, tension and lateral load testing of the piles.
- 11.1.2 Driving tests pile to the indicated depth and/or indicated resistance.
- 11.1.3 Keep a penetration record of tests piles and anchor piles in accordance with requirements set forth in Clause 10.
- 11.1.4 Pile load testing shall be carried out based on standard adopted and approved by Owner.
- 11.1.5 Upon completion of the preliminary pile load testing, the Bidder shall cut off the preliminary test piles, as directed by the Owner and dispose of the cut-off portion.
- 11.1.6 Cast in place concrete top piles shall be cured for a minimum of fourteen (14) days before applying load on test pile an anchor piles.

11.2 METHOD OF TEST

11.2.1 General Test Provisions

- 11.2.1.1 Perform test to the tests load indicated on the approved Bidder's drawings or as specified herein. Test equipment for the compressive load tests shall be capable of applying the required test load. Test equipment for the lateral and tensile load tests shall be capable of applying a tests load equal to 300% of the tests design loads shown on the approved Bidder's drawings.
- 11.2.1.2 Care shall be taken to ensure that the load applied by the hydraulic jack is coaxial with the pile.
- 11.2.1.3 Maintain test continuously during test period. If test is stopped before completion, and the load is wholly or partially removed from the test pile due to improper or insufficient loading, yield of support or connections, malfunctioning of measuring and testing equipment or for other mechanical reasons or for reasons for which the Bidder is responsible, the Owner shall order the

tests abandoned and replaced by a new test on another pile at an adjacent location and without additional cost.

11.2.1.4 Prior the driving each preliminary test pile, the Bidder shall drill a bore hole located between 1 and 2 meters from each test pile. The boring shall consist of continuous sampling with Standard Penetration Test (SPT). The boring shall penetrate with continuous sampling. A Dutch Cone penetration tests shall also be performed at each boring location.

11.2.1.5 Before commencing any tests the Bidder shall submit for approval full details of his proposals including the datum beam arrangements and location of support. Working piles shall not be used as tension reaction piles.

11.2.1.6 Sufficient number of anchor piles shall be driven to prevent excessive movement of anchor piles. All devices for measuring the settlement of piles shall be sheltered, and rigidly attached to a firm and independent support driven not less than 2.5 m from the nearest point of the test pile or anchor piles.

11.2.1.7 The calibration load cell shall be used for the load measurement.

11.2.2 Compressive Load Test

11.2.2.1 Perform compressive load tests on individual piles in accordance with applicable requirements of ASTM D1143.

11.2.2.2 Hydraulic jack and anchored reaction member method shall be used to apply the test load. Jacking against production piles acting in tension shall not be permitted.

11.2.2.3 Prior to the start of production pile driving, compressive test to failure shall be performed on individual test piles driven for testing purposes. The location of the tests piles shall be as indicated on the Bidder's drawings.

11.2.2.4 Compressive tests shall be performed on individual production piles selected by the Owner. The approximate location and number of the tests shall be as indicated on the approved Bidder's drawings.

11.2.2.5 Test loads for piles tested to failure shall be applied in accordance with the procedure for loading in excess of the standard load test, as specified in ASTM D1143. Test loads on production piles shall be applied in accordance

with the standard loading procedure, as specified in ASTM D1143. The preliminary load tests to failure shall include the cyclical load test of ASTM D1143. Production pile load tests shall include the cyclic load tests only when indicated on the approved Bidder's drawings.

- 11.2.2.6 For the preliminary compression tests, loading shall be increased as specified in ASTM D1143 until failure or when 300% of the test design load is reached. Loading for the production piling tests shall be increased as specified in ASTM D1143.
- 11.2.2.7 Monitor pile head deflection using a minimum of 4 dial gages, mounted to an independent frame and located radially about the pile. The dial gages shall have valid calibration.
- 11.2.2.8 Failure shall be defined as the load at which vertical deflection of the pile head exceeds 50 mm.
- 11.2.3 Tensile Load Test
 - 11.2.3.1 Perform tensile load tests on individual piles in accordance with applicable requirements of ASTM D3689
 - 11.2.3.2 Test loads shall be applied in accordance with the procedure for loading in excess of 200 % of pile design uplift load, as specified in ASTM D3689. The tests shall include the cyclical load test of ASTM D3689.
 - 11.2.3.3 Loading shall be increased until failure occurs.
 - 11.2.3.4 Tension tests shall be performed on individual piles driven for testing purposes. The location of the tests piles shall be as indicated on the approved Bidder's drawings.
 - 11.2.3.5 Monitor pile head deflection using a minimum of 4 dial gages mounted to an independent frame and located radially about the pile head.
 - 11.2.3.6 Failure shall be defined as the load at which uplift continues to increase without any further increase in load.
 - 11.2.3.7 No concrete shall be placed for pile caps until installation of reinforcement, tell tales and tension connection is complete. Design of tension connection shall be the Bidder's responsibility.

11.2.4 Lateral Load Test

- 11.2.4.1 Perform lateral load tests on single piles in accordance with the applicable requirements of ASTM D3966. Test shall be performed for fixed head and free head conditions.
- 11.2.4.2 Test load shall be applied in accordance with the procedure for loading in excess of the standard load test load, with cyclic loading, as specified in ASTM D3966.
- 11.2.4.3 Loading shall be increased until failure occurs. Failure shall be defined as the load at which horizontal deflection continues to increase without any further increase in load, or any lateral movement at design soil top elevation exceeding 6 mm.
- 11.2.4.4 Lateral load tests shall be performed on separate piles driven for testing purposes. The location of the test piles shall be as indicated on the design approved drawings.
- 11.2.4.5 Monitor pile butt movement and rotation of the head of the tests piles.

11.3 TEST EQUIPMENT AND MATERIALS

- 11.3.1. Provide all equipment and materials required for the tests and remove all such equipment and materials at the conclusion of tests.
- 11.3.2. Submit drawings of test arrangements, including full data for equipment and materials, for review and the Owner's approval, in accordance with the requirements set forth in Part 4, Section 2.
- 11.3.3. Test Result: Submit test results completed with tabulation of all test data to the Owner immediately after conclusion of test. Data shall include, as a minimum, calibration result of all test loading equipment, all dial gauge readings at each increment and decrement of load, time duration of test with time started and time completed and any unusual occurrences during testing.

12. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall be in accordance with the requirements of Part 4, Subsection 2.7 and this Technical Specification.

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